In class, we’ve gone through how AI has enabled the development of smart cities and their application in transportation.

First we’ve gone through the building blocks of smart cities such as the core technologies (IoT sensor networks, 5G connectivity, edge computing, AI/ML Platforms, and blockchains for security), infrastructure layers (smart roads & signals, connected public transit, intelligent parking systems, environmental monitors), and integration frameworks (data collection → analysis → action, real-time response systems, cross-system communication).

Then we went through AI-powered smart transportation via predictive traffic analysis, like weather impact modeling and event-based adjustments; computer vision, such as pedestrian tracking and incident recognition; and decision support through autonomous navigation, safety systems, and emergency responses. Along with the real-world impact

We’ve also gone through a series of tasks that would aid in the formation of a smart city, such as Intelligent traffic management through real-time monitoring, smart control systems, and incident response systems; connected vehicle technologies, like V2X communications (which include V2V (vehicle-to-vehicle), V2I (vehicle-to-infrastructure) and V2P (vehicle-to-pedestrian)), the standard security and privacy; next-generation public transit, with real-time operations, AI passenger experience, and smart infrastructure; smart parking technologies like detection and guidance, user experience, system benefits; urban environmental intelligence through monitoring systems, real-time analysis, and response mechanisms; and data analytics and management with data architectures, analytics tools, visualization, reporting, security, and privacy.

We’ve also gone through mobility as a service, urban logistics, social impact, economic impact, policy and regulation, emerging technologies and trends, and global smart city implementations.

With nearly every slide, they showed the impact of some services and the benefits of others. Such as how urban environmental intelligence can reduce high-pollution events by 25%, or how next-generation public transit has a 45% improvement in on-time performance, or how AI powered smart transportation improves incident responses by 40%, and how the economic impact be it direct or indirect include benefits like 20-30% on maintenance costs, 35% reduction in system downtime, mobility services, digital transformation, and more. All of this is to highlight how meaningful and important each of these technologies and implementations is.

All in all, there are many ways AI can be integrated into urban systems, leading to connected infrastructure, data-driven decision-making, and sustainable solutions.